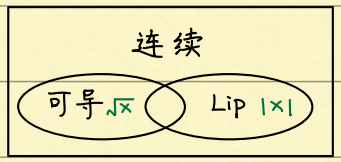
Lipschitz Ctn $|f(y)-f(x)| \le L_0 \|y-x\|$

L_0 越小越难，找最小的 L_0

e.g. $f=C$ $L_0=0$

$f=\sqrt{x}$ 不满足 Lip cnt，因为激增



Lip Ctn 为更强的连续

$L_0=\infty$ 即普通连续

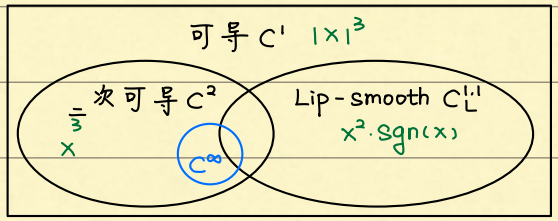
Lipschitz Smooth $\|\nabla f(y)-\nabla f(x)\| \le L\|y-x\|$, $f(x) \in C_L^{1,1}$

L 越小越难，找最小的 L

e.g. Lip $\frac{1}{2}x^TAx+\langle b, x\rangle$, $L=\|A\|$ 二次函数相似度

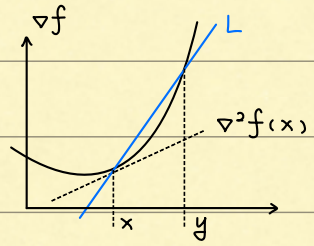
$\nabla f(x)=Ax+b$, $\|\nabla f(y)-\nabla f(x)\| = \|A(y-x)\| \le \|A\|\cdot\|y-x\|$

$\langle b, x\rangle$, $L=0$



2nd Lemma 若 $f \in C_L^{1,1}$，则 $\|\nabla^2 f\| \le L$

当 $f \in C^\infty$, $L=\|\nabla^2 f(x)\|$



Pf：Newton Lebniz：

$$\nabla f(x+\alpha h)-\nabla f(x) = \int_0^\alpha \nabla^2 f(x+th)\,dt$$

So $\|\sim\| \le \alpha L\|h\|$, as $\alpha \to 0$

Descent Lemma

若 $f \in C_L^{1,1}$, $|f(y)-f(x)-\langle\nabla f(x), y-x\rangle| \le \frac{L}{2}\|y-x\|^2$

若 $f \in C_L^{1,1}$ 且 convex

$f(y) \le f(x)+\langle\nabla f(x), y-x\rangle+\frac{L}{2}\|y-x\|^2$ 二次函数取等

Pf : Newton Lebniz : $f(y)-f(x)=\int_0^1 \langle \nabla f(x+t(y-x)),\ y-x\rangle dt$

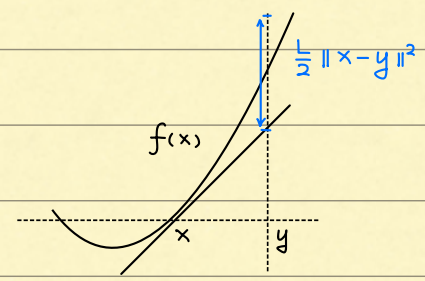
Then

$|f(y)-f(x)-\langle \nabla f(x),\ y-x\rangle|$

$=\left|\int_0^1 \langle \nabla f(x+t(y-x))-\langle \nabla f(x),\ y-x\rangle,\ y-x\rangle dt\right|$

$\leq \int_0^1 |\langle\ \rangle| dt \leq \int_0^1 \|\cdot\|\ \|y-x\| dt \leq \int_0^1 tL\|y-x\|^2 dt = \frac{L}{2}\|y-x\|^2$

convex func, $f(y)\geq f(x)+\langle \nabla f(x),\ y-x\rangle$



Strong Convex

$f(\alpha x+\beta y)\leq \alpha f(x)+\beta f(y)-\frac{\sigma}{2}\alpha\beta\|x-y\|^2$

等价于 $f(x)-\frac{\sigma}{2}\|x\|^2$ 凸

Pf : $g(x)=f(x)-\frac{\sigma}{2}\|x\|^2$ 凸

即 $g(\alpha x+\beta y)\leq \alpha g(x)+\beta g(x)$

$f(\alpha x+\beta y)-\frac{\sigma}{2}\|\alpha x+\beta y\|^2$

$\leq \alpha f(x)+\beta f(y)-\frac{\sigma}{2}\alpha\|x\|^2-\frac{\sigma}{2}\beta\|y\|^2$

其中 $\|\alpha x+\beta y\|^2-\alpha\|x\|^2-\beta\|y\|^2$

$=\alpha^2\|x\|^2+\beta^2\|y\|^2+2\alpha\beta\langle x,y\rangle-\alpha\|x\|^2-\beta\|y\|^2$

$=\alpha(\alpha-1)\|x\|^2+\beta(\beta-1)\|y\|^2+2\alpha\beta\langle x,y\rangle$

$=-\alpha\beta\|x-y\|^2$, 因为 $\alpha+\beta=1$

σ-strong convex, 1st 充要条件

$f(y)\geq f(x)+\langle \nabla f(x),\ y-x\rangle+\frac{\sigma}{2}\|x-y\|^2$

Strong Convex 解唯一

L-smooth, $\frac{1}{L}$-strong convex

σ-strong convex, $\frac{1}{\sigma}$-smooth